

Band Selection for Hyperspectral Land-Cover Classification: Comparing Classical and Learnable Feature Selection on the Salinas Scene

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Abstract

Hyperspectral sensors record hundreds of narrow spectral bands per pixel, but not all bands are equally informative for land-cover classification, and using all of them is costly and, under limited sample sizes, can actively hurt classifier performance. We study how many and which spectral bands are actually necessary for accurate per-pixel classification on the Salinas scene (204 bands, 16 classes, 54,129 labeled pixels), comparing two classical feature selection methods, LASSO and mRMR, against a learnable, jointly-trained selection method, Stochastic Gates (STG). We empirically justify a subset size of $k = 30$ bands via a sweep across candidate sizes, and evaluate all three selection methods using a shared, independently-tuned gradient-boosted tree classifier so that differences in downstream accuracy can be attributed to the selection method rather than to model choice. We additionally train a spectral 1D-CNN as a structurally different second model class. At $k = 30$, LASSO and STG-selected bands achieve test macro F1 of 0.975 and 0.974 respectively (99.5% and 99.4% of the full-band score), and a paired bootstrap test confirms these two are not significantly different from one another ($p = 0.298$), while mRMR trails significantly behind both ($p < 0.001$). A SHAP-based post-hoc attribution analysis of the full-band tree model agrees more with STG’s and LASSO’s selections than with mRMR’s, suggesting that selection methods which incorporate model feedback during selection align better with what a trained classifier actually relies on than a purely information-theoretic filter method does.

1 Introduction

Hyperspectral imaging captures a full spectral signature, typically over one hundred narrow, contiguous wavelength bands, for every pixel in a scene, rather than the three broad channels of an ordinary photograph. This makes hyperspectral data

attractive for tasks such as land-cover and crop classification, where different materials leave distinct, exploitable signatures across the spectrum. It also creates a concrete statistical problem: many of those bands are highly correlated with their neighbours, some are dominated by sensor or atmospheric noise, and classifier performance can degrade as the number of features grows relative to the number of labeled samples available (Hughes, 1968). This motivates band selection as a problem in its own right, independent of any specific downstream task: which of the available bands are actually necessary to classify a pixel correctly, and does the answer depend on how the selection is performed?

We study this question on the Salinas scene, a widely used hyperspectral benchmark of an agricultural valley in California, comparing three band selection methods that represent distinct methodological philosophies: LASSO, an embedded method that selects features as a side effect of fitting a sparsity-penalized linear model (Tibshirani, 1996); mRMR, a filter method that scores features by mutual information with the label while explicitly penalizing redundancy between selected features (Peng et al., 2005); and Stochastic Gates (STG), a learnable selection mechanism that trains a differentiable gate per feature jointly with a neural classifier, using a continuous relaxation of the ℓ_0 norm as a sparsity penalty (Yamada et al., 2020). Because STG’s mechanism requires backpropagation through the gates, it is inherently coupled to a neural network classifier, which lets us also compare a gradient-boosted tree against a spectral 1D-CNN as two structurally different model classes on this data.

Our contributions are: (1) an empirically justified choice of subset size k , rather than an assumed round number, obtained from a validation sweep; (2) a controlled comparison of the three selection methods using a single, independently-tuned down-

stream classifier, isolating the effect of the selection method from the effect of model choice; (3) bootstrap standard errors and paired significance tests for all reported comparisons, rather than point estimates alone; (4) a comparison between a tabular tree model and a spectral-locality-aware 1D-CNN on the same data; and (5) a SHAP-based attribution analysis of the trained tree model, cross-checked against all three selection methods’ outputs.

2 Related Work

Band selection for hyperspectral imagery has a long history in the remote sensing literature, motivated directly by the curse-of-dimensionality argument formalized by Hughes (1968): with a fixed number of labeled samples, classifier accuracy can decrease past a certain feature dimensionality, rather than merely plateauing (Sun and Du, 2019). Classical approaches split broadly into filter methods, which score features independently of any classifier (e.g. mutual-information-based ranking with redundancy penalization, as in mRMR; Peng et al., 2005), and embedded methods, which obtain a selection as a by-product of fitting a sparsity-regularized model such as LASSO (Tibshirani, 1996). More recently, learnable selection layers that are trained jointly with a downstream neural network have been proposed as a differentiable alternative to both; we use Stochastic Gates (Yamada et al., 2020), which relaxes the discrete, non-differentiable ℓ_0 feature-count penalty into a continuous, trainable one via per-feature Gaussian noise and clipping.

On the modeling side, gradient-boosted trees remain strong, widely used baselines for tabular data of this kind, and there is direct empirical evidence that they continue to outperform neural networks on many tabular tasks even when the networks are reasonably well tuned (Grinsztajn et al., 2022). For hyperspectral data specifically, convolutional architectures that operate directly on the raw per-pixel spectrum, rather than on hand-engineered features, have been proposed as a way to exploit the fact that neighbouring wavelength bands are physically correlated; we follow the architecture described by Hu et al. (2015), consisting of one convolutional layer, one max-pooling layer, one fully-connected hidden layer, and an output layer.

For model interpretation, we use SHAP (Lundberg and Lee, 2017), a unified, game-theoretic feature attribution method, to ask a question distinct from selection: not which bands a selection method

chose in advance, but which bands a model trained on all available bands ends up actually relying on.

3 Data

We use the Salinas scene, an AVIRIS airborne hyperspectral image of an agricultural valley in California (National Aeronautics and Space Administration, Jet Propulsion Laboratory, 2025), at 512×217 pixels with 3.7 m spatial resolution. The scene ships with 224 spectral bands, of which 204 remain after the standard removal of water-absorption bands, and 16 ground-truth land-cover classes (vegetable, soil, and vineyard fields). Of the 111,104 pixels in the scene, 56,975 are unlabeled background and are excluded, leaving 54,129 labeled pixels, an overall samples-to-feature ratio of roughly 265:1.

Preprocessing. Each band is z-score standardized, with the scaler fit only on the training split to avoid leaking test statistics into training. We use a stratified 64/16/20 train/validation/test split (34,642 / 8,661 / 10,826 pixels). One limitation is worth stating directly: hyperspectral pixels are spatially autocorrelated, so a random per-pixel split can place near-duplicate neighbouring pixels on both sides of the split, which may inflate reported accuracy relative to a stricter spatial (block-based) split. We accept this limitation in order to remain comparable with the majority of published work on this scene, which typically also uses random splits, and note it explicitly here rather than presenting results as free of this effect.

4 Methods

4.1 Baseline Classifier

We use a histogram-based gradient-boosted tree classifier (scikit-learn’s `HistGradientBoostingClassifier`) as our primary tabular model, chosen over a support vector machine for computational reasons: an RBF-kernel SVM scales poorly (roughly $O(n^2)$ to $O(n^3)$) at our training set size of 34,642 rows, whereas gradient-boosted trees remain efficient at this scale. Hyperparameters (`max_iter`, `learning_rate`, `max_depth`) are tuned via grid search with 3-fold cross-validation, optimizing macro F1 to account for class imbalance (the smallest class comprises under 2% of labeled pixels, the largest over 20%). The selected configuration (`learning_rate=0.05`, `max_depth=6`,

max_iter=400) was confirmed not to sit at a grid boundary by widening the search range and re-running; performance did not improve further at higher max_iter.

4.2 Feature Selection Methods

LASSO. We fit an ℓ_1 -penalized logistic classifier via stochastic gradient descent (SGDClassifier, log_loss, ℓ_1 penalty, early stopping), rather than the more common liblinear or saga solvers, which we found prohibitively slow at this feature-and-sample scale (single fits ranging from 6 to over 150 seconds; a full cross-validated search did not complete in reasonable time). The regularization strength is selected via grid search over $\{10^{-7}, \dots, 10^{-2}\}$ using 3-fold cross-validated macro F1; the chosen value ($\alpha = 10^{-6}$) was confirmed to be an interior optimum rather than a boundary artifact. Bands are ranked by the ℓ_2 norm of their per-class coefficients.

mRMR. We implement the standard minimum-Redundancy-Maximum-Relevance criterion (Peng et al., 2005): relevance is estimated via mutual information between each band and the label, redundancy via absolute Pearson correlation between candidate and already-selected bands, and bands are added greedily to maximize relevance minus mean redundancy with the current selection.

Stochastic Gates (STG). Following Yamada et al. (2020), we attach one learnable gate per input band to a small feedforward classifier, using Captum’s GaussianStochasticGates implementation (Kokhlikyan et al., 2020). Each gate is a clipped, noise-perturbed version of a learnable parameter μ_k ; the noise is active only during training, making the gate a continuous relaxation of a discrete on/off switch. The joint training objective adds a differentiable approximation of the expected number of open gates to the classification loss, scaled by a regularization weight λ . We tune the classifier architecture (hidden layer width $\in \{32, 64, 128\}$) once, with $\lambda = 0$, using early stopping on validation macro F1; width 128 was selected, though the gain over width 64 was marginal. We then sweep $\lambda \in \{0.003, \dots, 0.01\}$ with the architecture fixed, and select the value whose resulting number of open gates is closest to our target subset size (Section 4.3).

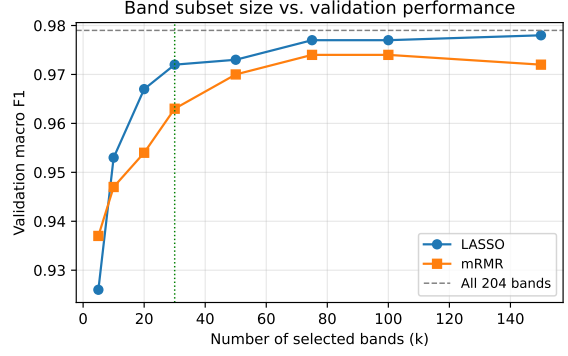


Figure 1: Validation macro F1 as a function of selected band count, for LASSO and mRMR, against the all-band baseline. Performance plateaus near $k = 30$ (dotted line).

4.3 Choosing the Subset Size k

Rather than assume a subset size, we sweep $k \in \{5, 10, 20, 30, 50, 75, 100, 150\}$ for both LASSO and mRMR, retraining the (already-tuned, not re-tuned per subset) baseline tree at each k and recording validation macro F1. Figure 1 shows the result: performance rises sharply up to $k \approx 30$ and then proceeds to only increase marginally, with LASSO gaining only 0.006 macro F1 in total between $k = 30$ and $k = 150$ despite a five-fold increase in band count. We select $k = 30$ as the point past which additional bands yield diminishing returns: a five-fold increase to $k = 150$ improves LASSO’s validation macro F1 by only 0.006, while keeping k small enough to constitute a meaningful feature reduction and to permit λ -matching for STG at an equal subset size across all three methods.

4.4 Spectral 1D-CNN

To test whether a model that can exploit the ordering of the spectrum outperforms a tree that treats bands as an unordered tabular vector, we train a 1D convolutional network directly on the raw per-pixel spectrum, following the architecture of Hu et al. (2015): one convolutional layer, one max-pooling layer, one fully-connected hidden layer (fixed at 64 units), and an output layer. Kernel size $\in \{3, 5, 7\}$ and filter count $\in \{16, 32\}$ are tuned via grid search with early stopping on validation macro F1; the hidden layer width was fixed rather than swept to bound the size of the search grid. Because current PyTorch wheels are not published for Intel-based macOS, this model was trained with torch 2.2.2 in a dedicated environment; band-subset evaluation

of the CNN (i.e. retraining it on each selection method’s 30 bands) was not performed, due to the computational cost of the architecture search alone (approximately two hours on the available CPU-only hardware) – we report the CNN only on the full 204-band input and note this as an explicit scope limitation rather than an oversight.

4.5 Attribution

For the tree model trained on all 204 bands, we compute SHAP values (Lundberg and Lee, 2017) via TreeExplainer on a random sample of 500 test pixels, averaging absolute SHAP values across samples and classes to obtain a single importance score per band. We compare the resulting top-30 bands against each selection method’s top-30 bands using the Jaccard index.

4.6 Significance Testing

All reported test-set metrics are point estimates from a single fixed split and are subject to sampling variability. To quantify this, we compute a non-parametric bootstrap standard error and 95% confidence interval for each model’s test macro F1: the test set is resampled with replacement 10,000 times, macro F1 is recomputed on each resample, and the standard deviation and percentiles of the resulting distribution are reported. To test whether two models differ significantly, we use a paired bootstrap test: for each of the 10,000 resamples, both models’ macro F1 is computed on the *same* resampled indices, and the distribution of paired differences is used to obtain a two-sided p -value (the fraction of resampled differences with the opposite sign to the observed difference, doubled) and a 95% confidence interval on the difference itself. Pairing on identical resampled indices, rather than resampling each model independently, is what makes this a valid paired test rather than a comparison confounded by extra between-resample noise.

5 Results

5.1 Baseline and Selection Comparison

Table 1 reports test-set performance, with bootstrap standard errors, for the tuned tree classifier on all 204 bands and on each selection method’s 30 selected bands, using identical hyperparameters throughout so that differences reflect the selected bands rather than the model. All three selection methods substantially reduce the feature count (to 14.7% of the original bands) while retaining most

Band set	Test macro F1	SE
All 204 bands	0.980	0.001
LASSO-30	0.975	0.001
mRMR-30	0.964	0.002
STG-30	0.974	0.001

Table 1: Tree classifier test macro F1, all bands versus each method’s 30 selected bands, with bootstrap standard error (10,000 resamples).

Comparison	Diff.	p
All vs. LASSO-30	+0.0046	< 0.001
All vs. mRMR-30	+0.0155	< 0.001
All vs. STG-30	+0.0060	< 0.001
LASSO-30 vs. mRMR-30	+0.0109	< 0.001
LASSO-30 vs. STG-30	+0.0014	0.298
mRMR-30 vs. STG-30	−0.0095	< 0.001

Table 2: Paired bootstrap tests between band sets (test macro F1, 10,000 resamples). Only LASSO-30 vs. STG-30 is not significant at $\alpha = 0.05$.

of the full-band performance; LASSO and STG are close to one another (99.5% and 99.4% of full-band macro F1 respectively), while mRMR trails both (98.4%).

Paired bootstrap tests (Table 2) confirm that the full-band model significantly outperforms every 30-band subset ($p < 0.001$ in all three cases), which is expected given the larger feature count. More informatively, **LASSO-30 and STG-30 are not significantly different from one another** (difference 0.0014, $p = 0.298$, 95% CI $[-0.001, 0.004]$, an interval that includes zero), while mRMR-30 is significantly worse than both LASSO-30 ($p < 0.001$) and STG-30 ($p < 0.001$). This upgrades what would otherwise be an eyeballed comparison of similar-looking numbers into a supported claim: LASSO and STG select bands of statistically indistinguishable downstream quality on this task, and mRMR’s shortfall is real rather than noise.

The overlap between the three methods’ selected bands is limited: LASSO and mRMR agree on only 17.6% of their selected bands (Jaccard index), sharing a cluster of adjacent bands (indices 3–8 and 150–152) but diverging elsewhere. This indicates that embedded sparsity and information-theoretic redundancy minimization are picking up on genuinely different signals in this data, rather than converging on the same underlying structure.

A more nuanced result emerges from comparing STG’s own classifier to the tree evaluated on STG’s bands. STG’s jointly-trained neural classifier reaches only 0.952 validation macro F1 at its

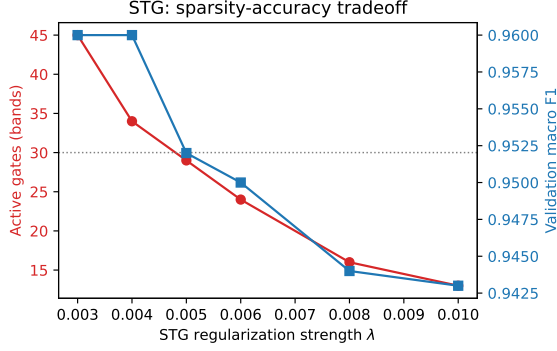


Figure 2: STG: number of active gates and validation macro F1 as a function of the regularization strength λ .

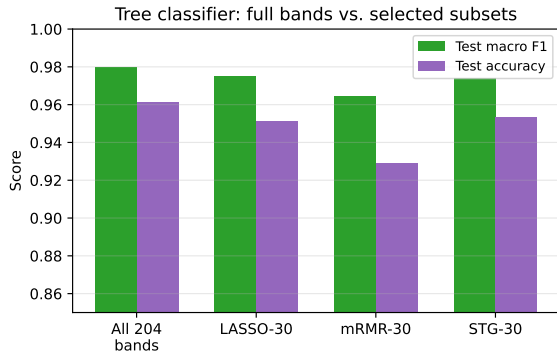


Figure 3: Tree classifier test macro F1 and accuracy, full band set versus each selection method’s 30-band subset.

matched subset size, well below the tree’s 0.970 on the identical band set. This gap does not indicate that STG selects poor bands; when those same bands are evaluated with the stronger tree classifier, performance is statistically indistinguishable from LASSO’s, as shown above. It indicates instead that STG’s own classifier, a small feedforward network, is the weaker component, consistent with broader findings that tree ensembles remain competitive with or superior to neural networks on tabular data of this kind (Grinsztajn et al., 2022). This distinction, between the quality of a method’s selected features and the quality of its own downstream classifier, is only visible because we evaluate all three methods’ selections with a shared, independent classifier rather than trusting each method’s native evaluation.

Figure 2 shows the STG sparsity-accuracy trade-off directly: validation macro F1 degrades gradually as λ increases and fewer gates remain open, without a sharp cliff, suggesting the selected bands are not highly dependent on features near the sparsity threshold.

5.2 Spectral CNN

The tuned spectral CNN (kernel size 7, 16 filters) reaches 0.981 test macro F1 and 0.957 test accuracy on all 204 bands, compared to the tree’s 0.980 macro F1 and 0.961 accuracy. A paired bootstrap test confirms these two structurally different model classes, a tabular tree treating bands as an unordered vector versus a CNN exploiting their spectral ordering, are **not significantly different** (difference -0.0014 in favor of the CNN, $p = 0.266$, 95% CI $[-0.004, 0.001]$, an interval that includes zero). An earlier CNN architecture that collapsed the entire spectral sequence to a single value per filter via global average pooling scored only 0.88–0.90 macro F1; correcting the architecture to preserve coarse positional information (following the pooling strategy implied by Hu et al., 2015, rather than an invented deeper design) recovered performance to parity with the tree. Both the tree and the CNN struggle on the same two classes (grapes-untrained and vineyard-untrained, known in the literature to be spectrally similar), which we take as evidence that this difficulty is a genuine property of the data rather than a model-specific artifact.

5.3 Attribution

The top-30 bands ranked by mean absolute SHAP value on the full-band tree model overlap with STG’s selection at Jaccard 0.429, with LASSO’s at 0.364, and with mRMR’s at only 0.111. This ordering is notable: the two methods that overlap most with what the trained model actually relies on (SHAP) are the two that incorporate feedback from a classifier during selection, LASSO via its regularized coefficients and STG via joint training with a classifier, while mRMR, a pure filter method that never references a classifier, aligns least. This is consistent with the interpretation that classifier-aware selection tends to agree more with post-hoc explanation of a trained classifier than classifier-agnostic filtering does, though we note this is a single-dataset observation and would need replication to generalize.

6 Discussion

Two results in combination are worth emphasizing. First, the choice of $k = 30$ was not assumed but derived from where the validation curve plateaus (Figure 1); at this operating point, over 85% of the original bands can be discarded while retaining 98.4–99.5% of full-band accuracy, depending

on the selection method. Second, the separation between STG’s selection quality and STG’s own classifier quality (Section 5.1) demonstrates why a matched-classifier comparison design matters: had we compared each method using only its own native classifier, we would have concluded STG selects poor bands, when in fact its bands are statistically indistinguishable in quality from LASSO’s, and the gap lies entirely in its own lightweight classifier.

The low agreement between mRMR and the other two methods, both with each other (Jaccard 0.176) and with the SHAP-derived importance ranking (Jaccard 0.111), together with its significantly lower downstream accuracy at matched k , suggests that purely information-theoretic filter selection, which never consults a trained classifier, may be systematically selecting a different, and on this dataset measurably less useful, notion of "importance" than methods that incorporate classifier feedback.

7 Limitations

Spatial split. As noted in Section 4, our random stratified split does not control for spatial autocorrelation between neighbouring pixels, which may inflate all reported accuracies somewhat relative to a stricter spatial-block split. We accept this for comparability with the majority of prior work on this scene, which uses the same convention.

CNN scope. The spectral CNN was evaluated only on the full 204-band input; band-subset evaluation of the CNN was not performed due to the computational cost of the architecture search on the available hardware. We cannot therefore state whether band selection benefits or harms the CNN in the way it does the tree, nor do we report a standalone bootstrap standard error for the CNN’s own test macro F1 in isolation, only its paired comparison against the tree.

Model scope. We deliberately restrict this work to a spectral-only, per-pixel framing (no neighbouring-pixel spatial context), to keep the band-selection question uncontaminated by spatial confounds. Current state-of-the-art hyperspectral classifiers are almost universally spatial-spectral, using a small image patch around each pixel rather than the pixel’s spectrum alone, and are reported to outperform spectral-only models by a substantial margin on this kind of data. Our results should

be read as characterizing band importance and selection-method behaviour within the spectral-only setting, not as competing with spatial-spectral state of the art.

Single-dataset scope. All results are reported on the Salinas scene alone; a cross-scene generalization test (e.g. on Pavia University, a different sensor and land-cover domain) was planned but not completed within the scope of this report.

8 Conclusion

We compared three band selection methods, LASSO, mRMR, and Stochastic Gates, for hyperspectral land-cover classification on the Salinas scene, using an empirically justified subset size, a shared downstream classifier to isolate the effect of selection method from model choice, and bootstrap-based significance testing rather than point estimates alone. All three methods retain the large majority of full-band accuracy using only 14.7% of the original bands; LASSO and STG are statistically indistinguishable from one another at matched subset size, while mRMR is significantly worse than both. A second, structurally different model class, a spectral 1D-CNN, was found to be statistically indistinguishable from the tuned tree despite exploiting spectral ordering rather than treating bands as an unordered feature vector. SHAP-based attribution of the full-band model agreed most with the two selection methods that incorporate classifier feedback during selection, and least with the purely filter-based method, suggesting that how a selection method obtains its ranking, not only how many features it keeps, meaningfully affects how well that ranking matches what a trained classifier actually uses.

Code availability. All code, configuration, and scripts used in this project are available at <https://github.com/LeonHauch/hui-band-selection>.

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